

Introduction to Contract Theory and Auctions

Interdependent Value Auctions

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On New Year's Eve of A.D. 192, the emperor Commodus was assassinated. This inaugurated the Year of the Five Emperors, A.D. 193. Pertinax was the first to claim the title. The Prætorian guard, which proclaimed him emperor, expected a generous donativum on his ascension, but they were disappointed. On the 28 of march of 193, Pertinax was at the palace when a contingent of three hundred soldiers from the Prætorian Guard rushed the gates, claiming that they had received only half of the promised pay. Pertinax attempted to reason with the insurgents but he was struck down by one of the soldiers.

After this bold move, the Prætorian Guard proceeded to sell off the entire Roman Empire by means of an auction. The winning bid was a promise of 25,000 sesterces per man to the Guard. The winner was the wealthy senator Didius Julianus, who was duly declared emperor. His reign lasted for only sixty five days, after which he suffered from what is perhaps the earliest and most extreme instance of the "winner's curse". He was beheaded.

His last words were "But what evil have I done? Whom have I killed?"

You are about to learn what he did wrong.

Relaxing the Assumption of Private Values

- Private Values: each bidder knows the value of the object to herself.
- We will now allow bidders to only have partial information regarding the value, in the form of a noisy signal.
- Bidders now have information that would, if known to other bidders, affect the value they assign to the object.
- This structure is called **interdependent values**.
- Suppose that Bidder i 's private information is summarized as the realization of a random variable $X_i \in [0, \omega_i]$, which is called his **signal** or type.
- Assume that the value of the object to bidder i , the random variable V_i can be expressed as a function

$$V_i = \nu_i(X_1, X_2, \dots, X_n)$$

where $\nu_i^n : \prod_i [0, \omega_i] \rightarrow \mathbb{R}_+$ is nondecreasing in all its variables and twice continuously differentiable. We will also assume that ν_i is strictly increasing in X_i .

- The specification above assumes that the value is completely determined by the signals, i.e., there is no remaining uncertainty.
 - Suppose that $V_i, i = 1, \dots, n$ denote the n unknown values to the bidders; $X_i, i = 1, \dots, n$ denote the n signals available to the bidders; and S a signal that is only available to the seller. Then define

$$\nu_i(x_1, \dots, x_n) = E[V_i | X_1 = x_1, X_2 = x_2, \dots, X_n = x_n]$$

- In either case, assume that $\nu_i(0, \dots, 0) = 0$ and that $E[V_i] < \infty$.
- We continue to assume that bidders are risk neutral and we will only consider standard auction formats: each bidder maximizes $V_i - p_i$ where p_i is the price paid.
 - We could use a generalization of definition 5.1 to interdependent values to consider other different instances of a sealed-bid auction.
- Private values: $\nu_i(\mathbf{X}) = X_i$.
- An important special case which is at the other end of the spectrum is a **Pure Common Value Auction**, in which all bidders assign the same value:

$$V = \nu(X_1, X_2, \dots, X_n)$$

Other assumptions and remarks

- A special case of the common value auction is when we start by specifying the distribution of V and then, assuming the conditional event $V = v$, we assume that bidder's signals X_i are independently distributed, i.e, **conditional independence** of X_i 's.
- Typically, it is also assumed that $E[X_i|V = v] = v$, i.e., X_i is an **unbiased estimator of the value**.

Things to be careful about:

- The interdependence of values complicates the decision problem of a bidder.
- Since the exact value of the object is unknown and depends on other bidder's signals, an *a priori* estimate may need to be revised as a result of the events taking place **during and after** the auction.
- Reason: events that happen during and after the auction convey **valuable information** about the signals of other bidders.
- One event that is crucial is the announcement of the bidder who has won.

The Winner's Curse

- Prior to the auction, the only information available to $i = 1$ is $X_1 = x$. Based on this, her estimate is $E[V|X_1 = x]$.
- Suppose the object is sold via a FPA and it is announced that 1 is the winner.
- If all bidders are symmetric and follow the same strategy β , what fact does being the winner reveal?
 - That the highest of the other $n - 1$ signals is less than x !
 - As a result:

$$E[V|X_1 = x, Y_1 < x] < E[V|X_1 = x]$$

- As a result, the announcement of winning leads to a decrease in the estimated value, i.e., **bad news!**
- A failure to foresee the above and take it into account is called **the winner's curse**.

The Winner's Curse

- Consider a pure common value auction, in which the bidder's signal is $X_i = V + \epsilon_i$.
- Suppose ϵ_i are i.i.d, and $E[\epsilon_i] = 0$.
- Then $E[X_i|V = v] = v$ and X_i is an unbiased estimator.
- However, the largest of n signals is not unbiased: using Jensen's inequality for convex functions, and with $\max$ being a convex function:

$$E[\max_i X_i | V = v] > \max_i E[X_i | V = v] = v.$$

- A bidder who does not take this into account and bids an amount $\beta(X_i)$ which is close to X_i would, upon winning, pay more than the estimated worth of the object.
- The winner's curse only arises if bidders do not calculate the value of winning correctly and overbid. Rational players playing an equilibrium strategy will not suffer from the winner's curse.

Non-equivalence of the SPA and the English Auction

- Under private values, we discussed that the SPA and the English Auction were equivalent.
- Why is this no longer the case with interdependent values?
 - In the English auction, active bidders get to know at which prices other bidders have dropped.
 - A SPA makes this information unavailable to the bidders by its very nature.
- Strategic equivalence of both auction formats is recovered if either:
 - 1 There are only two bidders.
 - 2 Values are private.

An Example with Independent Signals: FPA

$V = X_1 + X_2$, X_1, X_2 are i.i.d $U[0, 1]$

An Example with Independent Signals: SPA

$V = X_1 + X_2$, X_1, X_2 are i.i.d $U[0, 1]$

An Example with Independent Signals: Revenue Comparison

An example with Correlated Signals

$V = X_1 + X_2$, X_1, X_2 distributed as $f(x_1, x_2) = \frac{4}{5}(1 + x_1x_2)$, $(x_1, x_2) \in [0, 1]^2$.

Note that $f(x_2|x_1) = \frac{2(1+x_1x_2)}{2+x_1}$, $f(x_1|x_2) = \frac{2(1+x_1x_2)}{2+x_2}$.

- We will assume that all signals X_i are drawn from the same interval $[0, \omega]$ and that the distribution of the signals $f : [0, \omega]^n \rightarrow \mathbb{R}_+$ is a symmetric function, and that the signals are affiliated random variables.
- The valuation functions of all players will also be assumed to be symmetric in the following sense: for all i

$$V_i = \nu_i(\mathbf{X}) = u(X_i, \mathbf{X}_{-i})$$

and the function u , which is the same for all bidders, is symmetric in the last $N - 1$ components.

- From the perspective of a particular bidder, the signals of the other bidders can be interchanged without affecting the value.
- For example, with $N = 3$, $u(x, y, z) = u(x, z, y)$.

- Define the function

$$\nu(x, y) = E[V_1 | X_1 = x, Y_1 = y]$$

which is the expectation of the value to bidder 1, when the signal she receives is x and the highest signal among the other bidders, Y_1 , is y .

- Because of symmetry, this function is the same for all bidders.
 - Also ν is a non-decreasing in x and y , but we will assume more: ν is strictly increasing in x .
 - Since $u(\mathbf{0}) = 0$, then $\nu(0, 0) = 0$.
- $\nu(x, y)$ plays a very important role in the analysis we will carry out.

Proposition 9.1

Symmetric equilibrium strategies in a second-price auction are given by:

$$\beta^{II}(x) = \nu(x, x)$$

Proof.

Suppose all other bidders $j \neq 1$ follow the strategy $\beta = \beta^{II}$. Bidder 1's expected payoff when his signal is x and he bids an amount b is

$$U(b, x) = \int_0^{\beta^{-1}(b)} (\nu(x, y) - \beta(y)) g(y|x) dy = \int_0^{\beta^{-1}(b)} (\nu(x, y) - \nu(y, y)) g(y|x) dy$$

where $g(\cdot|x)$ is the pdf of $Y_1 = \max_{i \neq 1} X_i$ conditional on $X_1 = x$.

Since ν is increasing in the first argument, for all $y < x$, $\nu(x, y) - \nu(y, y) > 0$, and for all $y > x$, $\nu(x, y) - \nu(y, y) < 0$. Thus, U is maximized by choosing b so that $\beta^{-1}(b) = x$ or equivalently, $b = \beta(x)$. □

Derivation of the Equilibrium

- Suppose all bidders $j \neq 1$ follow the increasing and differentiable strategy β . Clearly, it is not optimal for bidder 1 to bid less than $\beta(0)$ or more than $\beta(\omega)$.
- Let $G(\cdot|x)$ denote the distribution of $Y_1 = \max_{i \neq 1} X_i$ conditional on $X_1 = x$ and let $g(\cdot|x)$ denote its pdf.
- The expected payoff to bidder 1, when her signal is x and she bids an amount $\beta(z)$ is

$$U(z, x) = \int_0^z (\nu(x, y) - \beta(z))g(y|x)dy = \int_0^z \nu(x, y)g(y|x)dy - \beta(z)G(y|x)$$

- The FOC is: $(\nu(x, z) - \beta(z))g(z|x) - \beta'(z)G(z|x) = 0$
- At a symmetric equilibrium, $z = x$ thus:

$$\beta'(x) = (\nu(x, x) - \beta(x)) \frac{g(x|x)}{G(x|x)}$$

Derivation of the Equilibrium in the FPA

- $\beta'(x) = (\nu(x, x) - \beta(x)) \frac{g(x|x)}{G(x|x)}$ is only a necessary condition.
- We must have that for all x , $\nu(x, x) - \beta(x) \geq 0$, since otherwise a bid of 0 would be better.
- Since by assumption, $\nu(0, 0) = 0$, this implies $\beta(0) = 0$.
- Note that the differential equation above can be written as:

$$\beta'(x) + \beta(x) \frac{g(x|x)}{G(x|x)} = \nu(x, x) \frac{g(x|x)}{G(x|x)}$$

- Trick with the integrating factor, define:

$$L(y|x) = \exp \left(- \int_y^x \frac{g(s|s)}{G(s|s)} ds \right)$$

- Notice that

$$\frac{d}{dy} L(y|x) = L(y|x) \frac{g(y|y)}{G(y|y)}$$

Derivation of the Equilibrium in the FPA

- Multiply the integrating factor $L(y|x)$ to both sides of the differential equation and simplify to get:

$$(\beta(y)L(y|x))' = \nu(y, y) \frac{g(y|y)}{G(y|y)} L(y|x) = \nu(y, y) dL(y|x)$$

since $\frac{d}{dy}L(y|x) = L(y|x) \frac{g(y|y)}{G(y|y)}$, and thus $dL(y|x) = L(y|x) \frac{g(y|y)}{G(y|y)} dy$.

- Solving the differential equation above with the boundary condition $\beta(0) = 0$:

$$\begin{aligned}\int_0^x (\beta(y)L(y|x))' dy &= \int_0^x \nu(y, y) dL(y|x) \\ \beta(x)L(x|x) - \beta(0)L(0|x) &= \int_0^x \nu(y, y) dL(y|x) \\ \beta(x) &= \int_0^x \nu(y, y) dL(y|x).\end{aligned}$$

Theorem 9.2

Symmetric Equilibrium strategies in the First-Price Auction are given by

$$\beta^I(x) = \int_0^x \nu(y, y) dL(y|x)$$

Proof

First note that $L(y|x)$ can be thought of as a distribution function with support $[0, x]$. Recall, from affiliation, that

$$\frac{g(s|s)}{G(s|s)} \geq \frac{g(s|0)}{G(s|0)}$$

Thus,

$$\begin{aligned} - \int_0^x \frac{g(s|s)}{G(s|s)} ds &\leq - \int_0^x \frac{g(s|0)}{G(s|0)} ds \\ &= - \int_0^x \frac{d}{ds} (\ln G(s|0)) ds \\ &= \ln G(0|0) - \ln G(x|0) \\ &= -\infty \end{aligned}$$

Applying the exponential function to both sides implies that for all x , $L(0|x) = 0$. Also $L(x|x) = 1$ and $L(\cdot|x)$ is non-decreasing. All of this implies $L(\cdot|x)$ is a distribution function.

Next, notice that if $x' > x$, then for all y $L(y|x') \leq L(y|x)$. Thus $L(\cdot|x')$ FOSD $L(\cdot|x)$. Since $\nu(y, y)$ is an increasing function of y , then $\beta = \beta^I$ is an increasing function of x .

Proof.

Consider a bidder who bids $\beta(z) = \beta(z)$ when his signal is x . The expected profit is

$$U(z, x) = \int_0^z (\nu(y, y) - \beta(z))g(y|x)dy$$

since β is increasing.

Differentiating with respect to z yields

$$\begin{aligned}\frac{\partial U}{\partial z} &= (\nu(x, z) - \beta(z))g(z|x) - \beta'(z)G(z|x) \\ &= G(z|x) \left[(\nu(x, z) - \beta(z)) \frac{g(z|x)}{G(z|x)} - \beta'(z) \right]\end{aligned}$$

If $z < x$, then since $\nu(x, z) > \nu(z, z)$ and because of affiliation,

$$\frac{g(z|x)}{G(z|x)} > \frac{g(z|z)}{G(z|z)}$$

we obtain that

$$\frac{\partial U}{\partial z} > G(z|x) \left[(\nu(z, z) - \beta(z)) \frac{g(z|z)}{G(z|z)} - \beta'(z) \right] = 0$$

Similarly, if $z > x$ then $\frac{\partial U}{\partial z} < 0$. Thus, $U(z, x)$ is maximized by choosing $z = x$.

Theorem 9.3

The expected revenue from a second-price auction is at least as great as the expected revenue from a first-price auction.